MAUVE-MUSE Paper II: the spatial progression of star formation suppression and residual ionised emission

Detailed manuscript-development report | 26 October 2026

Prepared for Rongjun Huang. This report follows the proposed seven-section paper, develops its scientific argument against the relevant literature, and provides a substantial draft Introduction. Section numbering below distinguishes the report from the proposed manuscript. The Introduction is intended as manuscript prose; the surrounding material records editorial decisions, evidence boundaries, and the remaining analysis required for submission.

Evidence status. The present assessment uses the live source and saved outputs of the three specified notebooks, six existing Paper II figures, the 20 October physical interpretation note, and primary literature accessed for this report. No notebook, FITS reduction, or science pipeline was rerun. Numerical descriptions of MAUVE results therefore refer to saved analyses, not an independent remeasurement or a new significance test. The proposed density and velocity-dispersion interpretations remain provisional. The earlier eight-findings review supplies background references; the present narrative follows the newer, more focused request.

Draft Introduction: 2054 words | 38 focused references

Begin with Section B for continuous manuscript prose. Sections C-F develop the paper structure, literature comparison, physical interpretation, and remaining analysis.

Contents

The report follows the proposed seven-section manuscript. The contents below refer to report sections; Section C maps these to the paper and its figures.

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A. The argument that should organise Paper II

A1. Central question and proposed title

The strongest organising question is: **How does the spatial decline of star formation connect to the disappearance, persistence, and changing excitation of detectable ionised emission across the MAUVE infall-stage populations?** This makes the SF and NSF results parts of one investigation. The rSFMS establishes the behaviour of the regions still selected as star forming. The SF/NSF/ND census then establishes where that selection ceases to occupy the observed disc. Finally, the NSF diagnostics describe the emission that remains detectable in those populations.

A suitable working title is:

MAUVE-MUSE: the spatial progression of star formation suppression and residual ionised emission in Virgo galaxies

An alternative with a stronger emphasis on the transition is **MAUVE-MUSE: from declining star formation to residual ionised emission along the Virgo stripping sequence**. The first is preferable at this stage because it accommodates differences between galaxies without implying that individual regions have been followed through time. Neither title requires a secure electron-density trend or a unique identification of HOLMES as the ionising source.

The principal observational statement can be developed as follows:

> *Across the adopted infall-stage populations, declining star formation is accompanied by different spatial changes in the emission census. The outer disc increasingly fails the adopted Balmer-detection requirements, whereas the inner disc retains a substantial component of detectable emission outside the adopted SF selection. The changing coverage of these components, together with the intensity of surviving SF and the declining brightness and equivalent width of NSF emission, provides a spatially resolved description of environmental suppression.*

This statement is deliberately about a measured distribution. The interpretation that outer gas removal and inner changes in the ionising-source mixture contribute to that distribution belongs in Section 6. An observed NSF location is not a record of a particular H II region becoming diffuse gas, and an ND location is not by itself proof that all gas has been removed.

A2. What is already established, and what this paper adds

Three strands of prior work provide the starting point. Virgo imaging and H I studies already establish truncated SF discs and outside-in environmental processing. Spatially resolved spectroscopy already establishes detectable LIER-like emission in regions with weak recent SF. Work on stripped galaxies, including MAUVE itself, already connects environmental disturbance with unusual optical line ratios and broader ionised-gas profiles. Paper II should acknowledge all three strands early and then bring them together through the same spatial census.

The most distinctive contribution is therefore the **joint description of SF intensity, SF/NSF/ND coverage, and the properties of the detectable NSF component as functions of local stellar surface density and radius across the adopted stages**. Its value is the connection between quantities that are often analysed separately. This is more specific, and more defensible, than claiming the first discovery of outside-in quenching, residual ionised gas, or local SF enhancement during stripping.

The exact combination has a credible claim to being a useful extension of earlier work, but the literature examined here does not justify an unrestricted priority claim. In particular, Boselli et al. (2020) already resolved the loss of H II regions in gas-stripped Virgo galaxies on approximately 50 pc scales, while Belfiore et al. (2017) established central and extended LIER configurations. These precedents leave room for a joint environmental analysis, rather than an assertion that either spatial outcome was previously unknown.

A3. The logical progression

The seven sections can be retained with only the numbering correction that observations and methods constitute Section 2. The progression is: establish the comparison and measurement definitions; describe surviving SF and the NGC 4654 exception; account for the spatial coverage lost from the SF category; characterise the residual detected emission; and interpret the connected measurements with a restricted physical model.

The NGC 4654 analysis should remain a focused subsection of Section 3. It establishes that local excess and broader suppression can coexist, but it should not displace the SF/NSF/ND transition as the principal result. Similarly, velocity dispersion and the [S II] doublet should test the interpretation without becoming conditions on which the entire paper depends.

B. Draft manuscript Section 1: Introduction

The decline of star formation in satellite galaxies is closely connected to the interaction between their gas reservoirs and the surrounding environment. In galaxy clusters, ram pressure exerted by the hot intracluster medium can displace and remove interstellar gas, while a reduction in subsequent gas supply can limit the replenishment of material consumed by star formation. These processes need not affect all gas phases or all locations within a galaxy simultaneously. Consequently, the relation between gas removal and star formation quenching depends on the structure of the galaxy, the distribution of its gas, and its environmental history (Gunn & Gott 1972; Cortese, Catinella & Smith 2021; Boselli, Fossati & Sun 2022). Understanding this connection requires measurements that resolve both the sites of ongoing star formation and the surrounding interstellar medium as the relative importance of young stellar populations declines.

The Virgo cluster provides a particularly useful environment in which to investigate these processes. Its proximity allows the distributions of atomic gas, recent star formation, and optical line emission to be compared within individual galactic discs. Hα imaging demonstrated that the reduced integrated star formation of many Virgo spirals is associated with truncated star-forming discs, while their inner regions can retain substantial activity (Koopmann & Kenney 2004). At finer spatial resolution, the Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE) showed that gas-stripped galaxies contain fewer H II regions and that this deficit becomes particularly pronounced outside the effective radius (Boselli et al. 2020). These observations establish that environmental suppression has a spatial structure that cannot be recovered from an integrated star formation rate alone. They also motivate a further question: what distinguishes the regions where star-forming emission is no longer detected from those where ionised emission remains visible after the contribution from recent star formation has weakened?

The atomic-gas morphology provides an independent basis for relating these spatial changes to environmental processing. The VLA Imaging of Virgo Spirals in Atomic Gas survey (VIVA) revealed a wide range of H I configurations, including compressed edges, asymmetric extensions, and severely truncated discs (Chung et al. 2009). Combining such information with projected phase space provides constraints on the likely stripping histories of Virgo galaxies (Yoon et al. 2017). However, a population classified by stripping state does not constitute a direct time sequence of otherwise identical galaxies. Differences in orbital geometry, gravitational interactions, and pre-existing structure can produce different responses at comparable environmental stages. A stage-based comparison is therefore most informative when it is interpreted as a comparison of spatially resolved populations, with the physical sequence constrained by additional gas and stellar information.

Gas removal can influence star formation through more than one route. In a retained molecular reservoir, reduced replenishment can lower the subsequent star formation rate without requiring an immediate change in the efficiency with which molecular gas forms stars. Conversely, changes in the molecular content, its dynamical state, or its conversion into stars can alter the relation between the gas reservoir and the observed star formation. A first-order correspondence between molecular-gas and star formation surface densities is observed in nearby discs, although the depletion time varies with environment and physical scale (Leroy et al. 2013). In Virgo, VERTICO measurements link environmental suppression to changes in both molecular-gas content and star formation efficiency (Brown et al. 2023). These results show why a decline in star formation alone cannot identify which part of the gas cycle has changed.

Integral-field spectroscopy enables this problem to be approached through the resolved star formation main sequence (rSFMS), which relates the local star formation rate surface density to the underlying stellar mass surface density. The relation has been established on approximately kiloparsec scales in CALIFA and extended to scales of order 100 pc with nearby MUSE surveys (Cano-Diaz et al. 2016; Erroz-Ferrer et al.

2019). It provides a reference against which local suppression and excess can be measured at comparable stellar surface density. However, its interpretation depends on which locations enter the measurement. A relation constructed from detected, SF-selected regions describes the surviving star-forming component, whereas an average across a larger disc footprint also depends on how much of that footprint contains such emission.

This distinction becomes especially relevant near the scale of individual star-forming complexes. Using PHANGS data, Pessa et al. (2021) demonstrated that the treatment of non-detections changes the measured rSFMS and its apparent dependence on spatial scale. Their work provides a particularly relevant comparison for MAUVE because the surveys probe similar physical scales with MUSE. It also shows that a comparison must account for spatial averaging, emission selection, and the quantity being fitted. Similarity in the overall rSFMS distribution can coexist with differences in the frequency and spatial distribution of star-forming locations. The rSFMS and the coverage of SF-selected emission therefore provide complementary information on the environmental response.

Observational studies also show that environmental processing need not produce uniform suppression at every location. The GAs Stripping Phenomena in galaxies with MUSE survey (GASP) has reported enhanced resolved star formation in selected stripping galaxies, while Virgo studies of molecular-gas efficiency do not find a universal enhancement on the windward side of stripped discs (Vulcani et al. 2020; Vollmer et al. 2012). The recent SAMI analysis of ram-pressure-affected cluster populations likewise finds spatial differences between asymmetric and truncated systems, with suppression most evident at low stellar surface density and large radius (Cakir et al. 2026). These findings are not described by a single expectation for all galaxies labelled as stripped. The measured outcome depends on the selected population, the stage of the interaction, and whether the comparison concerns local star formation, molecular-gas efficiency, or an integrated rate.

Individual Virgo galaxies illustrate the additional information supplied by spatial geometry. NGC 4654 has a well-established combination of tidal and hydrodynamical perturbations, and its compressed gas region has previously been associated with enhanced star formation efficiency (Vollmer 2003; Lizee et al. 2021). A directional optical excess in this galaxy can therefore be compared with an independently inferred gas-compression geometry, while allowing for a contribution from its tidal history. By contrast, the centrally concentrated activity in NGC 4424 occurs in a system for which a merger-related starburst and outflow have been identified alongside stripping (Boselli et al. 2018). These cases motivate a separation between the statistical behaviour of the stage populations and the physical interpretation of particular local enhancements.

The loss of SF-selected regions does not imply that optical line emission must disappear at the same locations or on the same timescale. Diffuse ionised gas can be maintained by photons escaping from H II regions, by hot low-mass evolved stars (HOLMES), or by additional sources of heating and ionisation. Spatially resolved MaNGA observations established that low-ionisation emission-line regions (LIERs) are widespread outside galactic nuclei and can be associated with old stellar populations (Belfiore et al. 2016). Central and extended LIER configurations further demonstrate that weak recent star formation can coexist with detectable ionised gas (Belfiore et al. 2017). Thus, the persistence of non-H II-like emission is already an established phenomenon. The environmental question is how its spatial prominence changes alongside the loss of star-forming emission and the growth of non-detections within the same observed disc.

Equivalent width and emission-line ratios offer complementary constraints on this residual emission. The equivalent width of Hα compares nebular emission with the underlying stellar continuum and therefore helps distinguish strong young-star-powered emission from weak emission associated with older populations (Cid Fernandes et al. 2011; Lacerda et al. 2018). Meanwhile, diffuse gas at low Hα surface

brightness commonly exhibits elevated low-ionisation forbidden-line-to-Balmer ratios (Zhang et al. 2017). Such behaviour means that weaker absolute line emission and larger diagnostic ratios can occur together. It also means that a location outside an adopted SF selection cannot be assigned a unique ionising source from its classification alone. The relation between surface brightness, equivalent width, and excitation must be examined explicitly.

The origin of diffuse emission remains sensitive to the local radiation field and gas geometry. In PHANGS-MUSE galaxies, Belfiore et al. (2022) examined a combination of leaking radiation from H II regions and ionisation by evolved stars. More recently, Baron et al. (2026) argued that the propagation and spectral hardening of leaked radiation can explain important properties of diffuse emission in actively star-forming discs. The latter study is currently a preprint and provides a relevant alternative to assigning all harder diffuse emission to HOLMES. Stellar-population and photoionisation calculations nevertheless show that post-AGB populations can produce weak Hα emission with LIER-like ratios (Byler et al. 2019). An environmentally processed sample spanning strong and weak recent SF can therefore help constrain where these explanations remain plausible, provided that photon production and gas absorption are considered separately.

Previous studies of stripped systems provide both guidance and important qualifications. Tomicic et al. (2021a) found no clear difference in the inferred integrated DIG contribution between their stripped and control samples. This result concerns an emission fraction and does not directly determine the fraction of disc area assigned to a non-SF category. In addition, diffuse gas in stripped tails can exhibit excitation requiring processes beyond ordinary star formation, potentially including shocks or mixing with the surrounding medium (Tomicic et al. 2021b). Those tail conditions need not apply to emission retained in the inner disc. Comparisons of UV and Hα in late-stage GASP galaxies also demonstrate that diffuse emission can be associated with a recent decline in star formation, highlighting the value of tracers with different temporal sensitivities (Tomicic et al. 2024).

Ionised-gas kinematics and density-sensitive line ratios provide additional tests of the physical interpretation. The association between broader emission lines and non-H II-like excitation is well established in resolved spectroscopic surveys (Law et al. 2021). However, an observed line width can contain contributions from unresolved velocity structure as well as local gas motions. Similarly, the [S II] doublet probes the density of the gas contributing to its emission and becomes weakly sensitive to density near its low-density limit (Kewley et al. 2019; Barnes et al. 2021). These diagnostics are therefore most useful when interpreted jointly with the spatial distribution and strength of the emission. They can test a proposed explanation without requiring every change in excitation to correspond to a unique change in turbulence or electron density.

MAUVE-MUSE was designed to investigate the gas cycle in nearby Virgo galaxies using spatially resolved observations of their stellar and gaseous components (Catinella et al. 2025). In Paper I, we used MAUVE observations to investigate the relation between local metallicity and star formation, and developed a resolved regulator framework for interpreting the balance between star formation and gas-supply variability (Huang et al. 2026). Complementary MAUVE work has already established environmental differences in the excitation and kinematics of the ionised medium relative to the PHANGS comparison sample (Brown et al. 2026). These studies motivate a more explicit connection between the behaviour of the remaining star-forming regions and the spatially resolved emission census across different stages of environmental processing.

In this work, we investigate that connection using the MAUVE-MUSE data at physical resolutions of order 100 pc. We first place the resolved SF distribution in the context of nearby MUSE measurements and examine its dependence on the adopted pre-peak, close-to-peak, and post-peak populations. We then

distinguish the intensity of surviving SF from the fractions of the usable footprint classified as SF, non-detected (ND), or detected but outside the adopted SF selection (NSF). This division allows us to ask where declining SF coverage is accompanied predominantly by a loss of detectable Balmer emission, and where detectable NSF emission remains prominent. Finally, we examine the surface brightness, line ratios, equivalent width, and supplementary gas diagnostics of the NSF population, and interpret their combined behaviour using a restricted model that links declining gas supply to changing ionising-source contributions. The aim is to establish how the different observed components are connected spatially, while distinguishing that observational connection from a unique evolutionary history or ionisation mechanism.

This paper is organised as follows. Section 2 describes the observations, derived quantities, stage assignments, and emission classifications. Section 3 presents the resolved SF comparison, the stage dependence of surviving SF, and the local excess in NGC 4654. Section 4 examines the spatial distribution of SF, NSF, and ND regions. Section 5 characterises the emission that remains detectable outside the SF selection. Section 6 develops the physical interpretation, compares it with previous work, and discusses the limitations of the model and observations. Section 7 presents our conclusions.

C. Detailed manuscript structure and figure plan

C1. Section 2: Observations, measurements, and comparison strategy

Purpose. Define the population to which every subsequent claim applies. The methods should explain the observational categories before attaching a physical interpretation to them.

The section should contain four compact subsections: the MAUVE sample and spatial resolution; stellar and emission-line measurements; environmental stages and SF/NSF/ND definitions; and statistical estimators and the external comparison. Include a sample table with product identifiers, physical galaxy names, inclination, stage provenance, usable coverage, and known nuclear activity or outflows. The saved main-notebook output contains 26 product groups, partitioned into 8 pre-peak, 5 close-to-peak, and 13 post-peak groups. NGC4567_8 is one weighting group. These are the current analysed product counts, not the design size of MAUVE and not necessarily a count of distinct physical galaxies.

The main notebook requires inclination below 80 degrees and finite stellar-continuum SNR_POSTFIT greater than 25 before classification. Its SF category requires a finite H II-associated SFR value, proxy EW(Hα) greater than 6 Angstrom, and positive, finite intrinsic Hα dispersion below 45 km/s. The upstream H II map carries BPT selection, with BPTMODE checked as both. ND means failure of the adopted detection requirement for Hα or Hβ. NSF is the remaining usable population. Consequently, NSF includes more than physically identified DIG, and ND does not mean that every emission line is absent. A scientifically complete methods table should also identify invalid or unavailable diagnostics within NSF instead of treating them as a pure excitation class.

The fraction and NSF profiles share the joint selection $7.5 \leq \log_{10}\Sigma_{\star} < 10.5$, with stellar surface density in solar masses per square kiloparsec, and $0 \leq R/R_e < 2$. The plotted radius and mass trends are nevertheless separate one-dimensional projections. Applying both bounds does not match the detailed mass distribution at every radius, or the radial distribution at every mass. The SF survivor-intensity panel uses its own 7.0-9.5 mass interval, and the overall rSFMS comparison has no additional radial cut. These distinctions must appear in captions and in the methods.

The primary population summaries should retain equal-galaxy weighting, supported by pooled-spaxel figures in an appendix. Whole-galaxy bootstrap bands describe uncertainty in the estimated stage trend. They do not turn native spaxels into independent observations, and a bin represented by one galaxy does not acquire population precision from a narrow bootstrap interval. Report the number of contributing galaxies by bin and retain the existing one-galaxy flags.

Figure 1: sample and coverage context, preferably an H I/stage overview with representative SF/NSF/ND maps. This figure is proposed; it is not already a product of the cited plotting notebook. Its role is to make the spatial categories and the meaning of the stages visible before the statistical comparisons.

C2. Section 3: The surviving star-forming component

3.1 The resolved SF distribution in the nearby-galaxy context. Present the PHANGS comparison as a calibration and interpretation step. The current plot shows a broadly overlapping locus and a pooled linear-mean upper-limit curve that lies near the published PHANGS relation over part of the stellar-density range. This is useful context, but it does not establish equality of complete two-dimensional distributions, fitted parameters, or intrinsic SF laws. A direct matched comparison remains a specific analysis task.

3.2 Stage dependence of SF intensity. The saved survivor-intensity profiles place pre-peak above close-to-peak and post-peak over much of the supported common density range. The ordering is not universal at the highest displayed bin. Describe the range in which the difference is supported rather than imposing three globally parallel rSFMS relations. Also distinguish the pooled median curves in the left panel from the usable-area-based conditional intensity in the right panel.

For a galaxy g in a fixed stellar-density interval, define the SF occupancy O_g and the SF-selected luminosity or SFR contribution per usable area T_g. The current stage estimator can then be expressed as

$$O_g = N_{\mathrm{SF},g}/N_{\mathrm{usable},g}, \qquad T_g = \sum_{i \in \mathrm{SF},g} \Sigma_{\mathrm{SFR},i}/N_{\mathrm{usable},g}.$$

$$O = \langle O_g \rangle_g, \qquad T = \langle T_g \rangle_g, \qquad I_{\mathrm{SF}} = T/O.$$

Thus T = O I_SF exactly for these definitions. This identity is the transition to Section 4: the SF-selected contribution can fall because the surviving SF is weaker, because SF occupies less usable area, or because both change. It is not the arithmetic mean of individual galaxies' conditional SF intensities, and T is not an estimate of all hidden or non-SF-contaminated star formation.

3.3 A local SF excess in NGC 4654. Show the candidate screen briefly, explain why NGC 4424 is a different physical case, and then present the NGC 4654 Fourier result. Retain the distinction between a non-zero asymmetry, a sector lying above the reference relation, and a causal attribution to RPS. An excess expressed in units of the notebook's width W must not be described as a number of Gaussian standard deviations: W is a custom corrected width, not a calibrated significance scale.

Figures 2-4: the rSFMS comparison; stage-dependent SF distribution and survivor intensity; and one compact NGC 4654 figure with residual map, fitted angular pattern, and the external H I geometry. Move the complete galaxy-by-galaxy screen, Fourier model variants, and detailed resampling results to an appendix.

Transition to Section 4. "The intensity of the surviving SF-selected component does not fully describe the change in the observable disc. We therefore examine how the relative coverage of SF, NSF, and ND regions varies with stellar surface density and galactocentric radius."

C3. Section 4: The spatial redistribution of emission categories

This is the empirical centre of the paper. Begin with the common denominator and the identity f_SF + f_NSF + f_ND = 1, then show the mass and radial profiles together. The result is not simply that the SF fraction decreases. It is that the accompanying increase in the other categories depends on position within the disc and on local stellar surface density.

The saved equal-galaxy radial figure supports a pre-peak SF fraction of approximately 0.4-0.6 across the displayed radii. In the close-to-peak population, the innermost bin is approximately 0.5, with lower SF fractions farther out and an increasing ND fraction. Post-peak SF coverage is approximately 0.25 in the inner few bins and falls below approximately 0.1 outside R_e, approaching very small values farther out. These are visual descriptions of the saved figure, not newly measured values with uncertainties. They refine the approximate values in the proposed narrative: "nearly absent" is more accurate than an exact zero immediately beyond R_e.

In the inner post-peak population, NSF is more prominent than in the pre-peak population, while ND dominates at larger radius. At the highest stellar surface densities, however, NSF already dominates some

pre-peak bins, so the manuscript should not claim a monotonic increase at every high-density location. The strongest reading is a position-dependent change in the relative incidence of the categories, with the largest useful contrasts evaluated where multiple galaxies contribute.

The comparison to the literature should be brief in the Results and developed in Section 6. Outside-in suppression agrees with the Virgo imaging and H I evidence. The extension is the simultaneous census of detectable NSF emission on the same usable footprint. The manuscript should say that the distributions are **consistent with different observable outcomes of declining SF**, rather than that particular SF regions have been observed transforming into NSF or ND regions.

Figure 5: the existing equal-galaxy mass and radial fraction figure, with contributing-galaxy counts and uncertainty bands. **Figure 6, if supported:** a compact joint radius-density comparison or matched-bin contrast showing that the apparent inner/outer distinction is not entirely a projection of differing stellar-density distributions. This second figure is a recommended addition, not an existing result.

C4. Section 5: The properties of detectable NSF emission

5.1 Absolute line surface brightness and relative line ratios. Place the brightness and ratio results next to one another. The saved profiles show substantially weaker NSF line emission in processed populations over much of the well-supported range, accompanied by larger values of several forbidden-line-to-Balmer ratios. Avoid asserting that all five line intensities strictly decline and all three BPT ratios strictly rise from pre-peak through post-peak in every bin. For example, the radial [S II]/Hα curves for close-to-peak and post-peak cross, and the highest-density close-to-peak points often represent one galaxy.

The plotted absolute quantity is corrected line luminosity surface density in erg/s/kpc^2. On the common six-line footprint, the notebook averages log intensities and individual log ratios with matched weights. The ratio trends are consequently linked algebraically to differential fading:

$$\Delta\langle \log_{10}(I_\ell/I_B)\rangle = \Delta\langle \log_{10} I_\ell\rangle - \Delta\langle \log_{10} I_B\rangle.$$

The physical information is which lines fade most strongly and how this depends on stage and position. The ratio and brightness panels are complementary representations of the same underlying measurements, rather than independent confirmations.

5.2 Hα equivalent width. The broadly lower EW in the later populations is consistent with a reduced nebular contribution relative to the stellar continuum. It is useful evidence in the connected narrative, but the NSF selection is already partly determined by EW. Demonstrate whether the trend persists for alternative classification choices and inspect the proxy continuum definition before interpreting absolute thresholds as a direct old-star fraction.

5.3 Kinematics and the [S II] doublet. Present intrinsic [N II] dispersion and the linear 6716/6731 ratio as supplementary constraints. Their current profiles do not establish a universal monotonic stage trend. Discuss the central dispersion differences, sparse high-density support, and the closeness of many sulfur ratios to the low-density limit. A null or weak density result remains scientifically useful: a change in excitation need not require a measurable change in this density diagnostic.

Figures 7-8: paired NSF brightness and BPT-ratio profiles; then EW, [N II] dispersion, and [S II] doublet profiles. If the latter two remain uncertain, retain EW in the main text and put the provisional diagnostics in the appendix. The principal argument remains intact.

C5. Section 6: A connected physical interpretation

Organise the Discussion by physical questions: why the outer and inner emission censuses differ; how declining gas supply can reduce the young-star contribution; why line ratios can rise while lines fade; what additional constraints EW, kinematics, and the doublet provide; and which alternative mechanisms remain allowed. The two-reservoir calculation belongs here as an interpretative construction. It should not be presented as a fit to the stage labels or as a direct measurement of stripping rates.

Make the distinction from Paper I explicit. Paper I connected metallicity-SFR behaviour to local supply and SF variability. Paper II uses related regulator bookkeeping to connect environmental gas processing to the changing visibility and excitation of ionised emission. The additional radiation and selection assumptions must be stated independently; success of the Paper I framework does not validate them automatically.

Figure 9: a schematic model prediction showing the young-star luminosity, the absorbed old-star contribution, total line luminosity, and line ratios for one clearly labelled illustrative parameter set. If this figure is later produced, use model time, not observational stage labels, on its time axis. A separate conceptual panel can indicate outer loss of detectability and inner persistence. No new model curve is generated in this report.

C6. Section 7: Conclusions

Use four or five numbered findings, ordered by their evidential strength. First describe the surviving SF distribution and its stage dependence. Second state the SF/NSF/ND spatial census. Third summarise the brightness, EW, and ratio behaviour of detectable NSF emission. Fourth describe the NGC 4654 directional excess as an individual case consistent with the known compression geometry. Finally state that a declining young-star contribution with persistent alternative ionisation is a viable interpretation, subject to gas absorption, source-mixture, and selection limitations. Include a density or dispersion conclusion only after its dedicated validation.

For the eventual manuscript, a useful allocation is approximately 1,800-2,200 words for the Introduction, 1,500-2,000 for Methods, 1,000-1,400 for Section 3, 900-1,200 for Section 4, 900-1,200 for Section 5, 1,800-2,400 for Discussion, and 350-500 for Conclusions. The draft Introduction above is intentionally expansive enough to supply material for that final editorial selection. The paper need not reproduce the report's full reference commentary or technical audit.

D. The comparison with previous work: what is established and what remains distinctive

D1. The rSFMS comparison is useful, but its present interpretation should be narrower

PHANGS is the most directly useful large MUSE comparison for this analysis, but it is neither the only survey approaching 100 pc resolution nor a purely field control. The MAD results of Erroz-Ferrer et al. (2019) provide another high-resolution precedent. The Pessa et al. (2021) sample includes Virgo members and overlaps the MAUVE comparison through NGC 4254, NGC 4321, and NGC 4535. Calling the comparison "MAUVE versus nearby star-forming PHANGS galaxies" avoids attributing every difference, or lack of difference, to a clean cluster-field contrast. The overlap should be listed and removed in an independent-control sensitivity check.

Pessa et al. used spatially sampled maps at several physical scales, included zero-valued non-detections in linear averaging, and fitted binned relations. Their emission treatment also includes a DIG correction. The current MAUVE notebook instead displays an upper-limit construction: stored ND limits are included, and NSF Hα is assigned an SFR-equivalent value under full young-star attribution. This is a deliberately generous observable proxy. It is not the same estimator as the PHANGS measurement, and it is not a censored-data likelihood. The distinction concerns both the meaning of an individual value and the way the values are averaged.

The current notebook's SF-only equal-galaxy individual-spaxel OLS has slope approximately 0.984 and intercept -10.034, while the displayed PHANGS relation has slope 1.05 and intercept -10.63 in the adopted units. These numbers do not support a statement of equal fitted relations: they refer to different selections and estimators, and slope and intercept cannot be compared independently over a broad mass range. The closer agreement of the displayed upper-limit mean curve over intermediate stellar density is a separate visual observation. Nor does overlap in a dense portion of the plot demonstrate equality of the full two-dimensional distributions.

There is nevertheless a valuable result to preserve. A familiar local SF locus can be populated by galaxies with very different fractions of their observed discs still meeting the SF criteria. The global comparison motivates the occupancy analysis rather than resolving the environmental question by itself. A manuscript-ready statement at the current evidence level is:

> *The MAUVE emission distributions overlap the region occupied by nearby PHANGS galaxies, while the stage-dependent measurements reveal differences within the MAUVE population. Because the treatment of non-SF emission and non-detections differs between the surveys, this comparison establishes broad consistency in the sampled parameter space rather than equality of the underlying rSFMS.*

The immediate refinement is to reproduce the same linear estimator on a common physical scale and stellar-density interval, document the treatment of missing and censored values, and compare independent galaxies. Matching effective resolution means matching the PSF and spatial averaging, not simply using native pixels of similar angular size. This report does not assume that the existing native-spaxel maps have already been convolved to a common 100 pc beam. Any censored-data analysis must also account for the way Hβ detectability, dust correction, and SF classification enter the selection; replacing every limit by zero or by its upper bound provides different descriptive brackets, not a complete statistical solution.

D2. The stage-dependent SF result extends an established environmental pattern

The relevant starting point is not that local SF suppression has never been demonstrated in clusters. Brown et al. (2023) already found a suppressed Virgo rSFMS on approximately 720 pc scales and connected the offset to molecular-gas content and efficiency. Their measurements also allow local enhancement in the outskirts of early-stage systems. Cakir et al. (2026) found stronger outer and low-density suppression in disturbed and truncated SAMI galaxies, with broadly comparable central SF across their classes. These studies support the spatial logic of the MAUVE interpretation while showing that the use of environmental categories alone is not the novel element.

MAUVE adds finer spatial discrimination and a consistent description of how detectable SF intensity, SF occupancy, and the alternative emission categories vary together. Differences from GASP should be presented as a comparison of sample selection, resolution, stage definition, and observable. A higher SFR at fixed stellar density is not the same measurement as higher SFR per unit molecular gas. Vulcani et al. (2020) and Vollmer et al. (2012) therefore need not disagree simply because one finds enhanced resolved SF in a selected stripping population and the other does not find a general increase in windward molecular-gas efficiency.

A useful discussion sentence is: "The coexistence of local enhancement with declining stage-averaged SF is consistent with a spatially heterogeneous response to gas compression and removal; the prevalence and amplitude of each response depend on the population and scale being measured." This formulation accommodates the NGC 4654 result without requiring it to explain the majority of the sample.

D3. NGC 4424 and NGC 4654: what the individual cases can establish

NGC 4424. The outflow reference is secure. Boselli et al. (2018) interpreted the galaxy in terms of a merger-associated central starburst, a starburst-driven ionised outflow, and stripping. Its ionised extension is directed differently from the long atomic-gas tail. The central SF excess is consequently not a clean tracer of a disc-plane, windward enhancement. It is appropriate to retain this galaxy in the candidate census and explain why it is excluded from the particular directional test. Excluding that test does not mean excluding the galaxy from all environmental interpretation.

NGC 4654. Vollmer (2003) modelled the joint influence of a tidal encounter, associated with NGC 4639, and ram pressure. This should be called a tidal encounter or flyby, not a flyby merger. Lizee et al. (2021) subsequently analysed the compressed gas region and its locally elevated molecular-gas efficiency. The new Fourier result is therefore an additional optical, spatially quantified diagnostic in a galaxy with an established mixed-interaction interpretation; it is not the first evidence for compression-associated SF in NGC 4654.

The saved Fourier output provides the following specific evidence. The MUSE-only first-harmonic phase is approximately 295.7 degrees, with bootstrap offsets of about -8.8 and +10.7 degrees. The adopted VIVA comparison is a coarse north-west sector represented by 315 degrees, giving a nominal separation of 19.3 degrees. The notebook explicitly records no measured uncertainty for that H I direction. Chung et al. (2009) describe compressed H I contours in the north-west and an extension to the south-east. The warranted statement is that the optical maximum lies in the same broad north-west sector. The calculation does not establish statistically exact alignment with a measured wind vector; the position-angle convention and the distinction between a compressed edge and tail direction should be shown in the figure.

The saved model-conditional wild-resampling probability is approximately 0.001 for the tested asymmetry. This concerns the fitted angular signal under that model and resampling scheme. It is not a probability that

RPS caused the signal. The selected peak sector has a positive median residual of approximately 0.127 W, whereas the full-galaxy median is approximately -0.208 W and the opposite sector is approximately -0.431 W. The fraction with residual $E > 2$ is approximately 0.112 in the peak sector, compared with approximately 0.070 for the pre-peak galaxy-average reference and approximately 0.070 for NGC 4654 as a whole. The pre-peak galaxies span a substantially wider range than their mean. These values describe a local excess superposed on a galaxy that is not globally elevated by the same measure.

Two checks are needed before presenting this as a final population result. First, the candidate-screen notebook uses different inclination and continuum-S/N rules for the pre-peak reference and the close/post-peak targets; the Fourier notebook also predates the current main-notebook selection. The statement that these are the only two candidates is therefore conditional on the exploratory screen. Second, selecting the peak direction from the data raises the expected contrast even for a control galaxy. Apply the same directional search to controls and include reference-relation uncertainty, spatial correlation, and model choice in the comparison. The measured asymmetry can remain useful even if the causal attribution stays mixed.

D4. Is the inner NSF increase novel?

The strongest answer is: **the components are established, while their connected environmental census is a defensible focus for Paper II.** The nearby literature prevents a claim that either outer disappearance or inner residual emission is newly discovered. It does not make the MAUVE combination redundant.

Closest precedent	What it already establishes	The specific MAUVE extension
Koopmann & Kenney (2004); Boselli et al. (2020)	Truncated Virgo SF discs and a reduced H II-region population, including at very fine spatial resolution.	Quantify the accompanying NSF and Balmer-ND coverage on the same usable footprint.
Belfiore et al. (2016, 2017); Lacerda et al. (2018)	Extended and central low-ionisation emission, weak-SF regions with detectable gas, and EW-dependent diffuse components.	Place that residual emission alongside the SF/ND census in a specifically environmentally classified Virgo population.
Tomicic et al. (2021a, 2021b)	DIG contributions and excitation in stripped galaxies, including disc-tail and stage-related comparisons.	Separate equal-galaxy area incidence from line-flux fractions and focus on retained inner-disc emission.
Poggianti et al. (2025)	A full GASP sample containing galaxies at different stages, including truncated gas discs and advanced loss of line-emitting gas.	Resolve the category distribution and its local-density dependence at MAUVE scales; do not claim the first stripping sequence.
Brown et al. (2026)	MAUVE-versus-PHANGS differences in excitation and ionised-gas kinematics.	Connect those diagnostics internally to the adopted stages, the SF/NSF/ND coverage, and the fading of absolute line emission.

The last comparison is especially important because it concerns the same survey. Brown et al. (2026), *MAUVE-MUSE: Ionization and Kinematic Signatures of Environmental Effects on Virgo Cluster Disks*, already analyse excitation and line widths in MAUVE relative to PHANGS. Paper II should state explicitly which questions are new and which measurements extend that work. It should not advertise enhanced non-SF ratios or broad emission in Virgo as discoveries independent of that publication. Its category

definitions also differ from the present notebook, so numerical fractions must not be transferred between the papers without reapplying a common selection.

There is no necessary contradiction with the absence of a clear stripped-control difference in the integrated DIG fraction reported by Tomicic et al. (2021a). Their quantity is based on emitted flux; the present f_NSF is based on spatial incidence under a different definition. An extensive but faint NSF component can occupy a large fraction of the usable area while contributing a modest fraction of total Hα luminosity. Likewise, a galaxy whose outer disc falls below detection can have an increased NSF fraction within the remaining observed region without an increased total mass of diffuse gas.

Two related distinctions should remain visible. "Inner" and "high stellar surface density" are correlated coordinates, not equivalent physical selections. Also, a fraction measured after a continuum-S/N cut describes the selected stellar disc, not automatically the entire geometrical area within 2 R_e. A change in the valid footprint can alter the denominator. These limitations can be checked directly and should not be used to discard the otherwise informative joint census.

D5. Fading NSF emission, EW, dispersion, and density

The expected comparison for declining EW is the WHAN, MaNGA LIER, and CALIFA DIG literature. Those studies establish why weak Hα relative to an old stellar continuum is compatible with a reduced young-star contribution. They do not imply that every MAUVE NSF spaxel is a retired region. The saved pre-peak NSF profiles include EW values well above commonly used retired-galaxy thresholds; the current NSF category can be entered through kinematics or BPT selection even when EW is substantial. The thresholds of Cid Fernandes et al. (2011), the regimes of Lacerda et al. (2018), and the notebook's 6 Angstrom cut serve different classifications and should not be substituted for one another.

Zhang et al. (2017) and the PHANGS DIG studies establish that lower surface brightness and elevated line ratios occur together outside a cluster-stage analysis. The MAUVE contribution is the dependence of those joint measurements on the environmental populations and location within the disc. An increase in a line ratio does not require an increase in the forbidden-line luminosity. For the current profiles, the physical question is whether Balmer emission fades more strongly than the forbidden lines on comparable, adequately detected samples.

The saved [N II] dispersion profiles are suggestive in the centres but do not define a monotonic sequence at all radii and stellar densities. The environmental interpretation must be compared with the general excitation-width association established by Law et al. (2021) and the direct MAUVE results of Brown et al. (2026). Using [N II] reduces the thermal contribution relative to hydrogen, but subtracting an instrumental width alone does not remove unresolved rotation, multiple components, or all thermal broadening. Moreover, the SF/NSF classification still uses Hα width; correlated or tied gas kinematics can create a selection dependence in an NSF [N II] comparison. The line is not an independent classification-free test merely because it is a different species.

The current [S II] profiles are even less suited to a single stage statement. Many ratios lie near approximately 1.4-1.46, where the density diagnostic is close to its low-density limit for ordinary nebular conditions. The saved central post-peak value is approximately 1.22, below the pre-peak value of approximately 1.29 and the close-to-peak value of approximately 1.38; other intervals have different ordering. These are visual approximate values, without a new uncertainty calculation. They rule out describing the displayed result as a universal progression towards larger 6716/6731 ratios and lower density. Near the limit, a density upper bound or consistency with low density is more meaningful than a sharply measured small density.

For a galaxy g, let A_g and B_g be the summed 6716 and 6731 intensities over N_g accepted pixels. The notebook's equal-galaxy estimator is

$$R_{\mathrm{eq}} = \frac{\sum_g A_g/N_g}{\sum_g B_g/N_g}.$$

This is a ratio of equally weighted galaxy-mean intensities. It is not the arithmetic mean of individual ratios or densities. Even after converting a well-measured ratio with an assumed temperature and atomic dataset, the result characterises the sulfur-emitting gas and need not equal the volume-average density of all ionised material. Kewley et al. (2019) and Barnes et al. (2021) provide the appropriate diagnostic context; a proposed stage-dependent density decrease requires its own evidence.

E. A physical discussion built from the 20 October model

E1. The useful explanatory content of the gas-supply calculation

The existing note contains a useful physical argument: environmental reduction of an atomic reservoir can reduce the supply to the molecular phase, and hence the luminosity supplied by young stars, while a more slowly varying stellar component remains. The general gas-regulator context is established by Lilly et al. (2013) and developed for a different MAUVE question in Paper I. In Paper II the calculation should be introduced as a minimal model of this one route to declining SF, with its additional radiation assumptions made explicit.

Let H(t) and M(t) denote the atomic and molecular reservoirs of a representative region. For constant coefficients, no external replenishment after the chosen onset, and no direct molecular stripping, the note's bookkeeping can be written as

$$\dot{H} = -aH, \qquad \dot{M} = H/\tau_{\rm conv} - bM, \qquad \psi = M/\tau_{\rm dep},$$

$$a = \tau_{\rm conv}^{-1} + \gamma_{\rm strip}, \qquad b = (1 - \mathcal{R} + \eta)/\tau_{\rm dep}.$$

Here psi is the SFR, R is the assumed promptly returned fraction, eta is the adopted mass-loading factor, and the treatment of recycling is part of the reservoir approximation. For a different from b,

$$H(t) = H_0 e^{-at}, \qquad M(t) = M_0 e^{-bt} + \frac{H_0}{\tau_{\rm conv}} \frac{e^{-at} - e^{-bt}}{b - a}.$$

For a = b, the second solution becomes

$$M(t) = e^{-bt}\left(M_0 + \frac{H_0 t}{\tau_{\rm conv}}\right).$$

The initial derivative makes the role of the initial condition transparent: molecular gas initially grows only if H_0/tau_conv exceeds b M_0. An initially balanced reservoir has zero initial derivative and subsequently declines as its atomic supply falls. An initially oversupplied molecular reservoir can rise before declining. This alternative is an illustrative initial condition, not a fitted observation of a stripping-induced burst. Increasing atomic removal at fixed initial reservoirs reduces the later supply; the baseline equations contain no compression-induced change in molecular formation or SF efficiency with which to generate a positive RPS response.

The distinction matters for NGC 4654. Its local excess may motivate a model with compression, transport, or altered molecular conversion, but it is not explained simply by increasing the stripping coefficient in the fading model. Likewise, the model's time origin is an assumed change in supply. It is not automatically the time of peak ram pressure, pericentre passage, or maximum observed SF. The three observational stage labels should not be placed at fitted times without an independent calibration.

The spatial factor c_HI in the note rescales the initial atomic reservoir while holding other chosen quantities fixed. It should remain separate from the temporal initial-condition experiment. It is not a measured local stripping rate, a local clock, or evidence that a specified part of the galaxy was compressed. The observation that different choices produce plausible spatial trends demonstrates the flexibility of the interpretation; it does not identify which choice occurs in the data.

Finally, neglecting direct molecular stripping is a simplifying limit. Evidence that molecular gas can be environmentally affected exists in Virgo and elsewhere, including Boselli et al. (2014). The baseline also omits changes in depletion time, radial inflow, and a time-dependent wind. These omissions are acceptable for an explanatory model if its purpose is stated clearly and the unmodelled channels remain part of the discussion.

E2. Why weaker lines and larger ratios can occur together

Let Y be the Hα luminosity surface density powered by the young-star contribution and L_o the Hα surface density powered by an alternative, relatively persistent component. The subscript o can denote an evolved-star component in the baseline interpretation; it should not imply that the data uniquely identify that component. If both components emit a forbidden line with fixed linear line ratios r_y and r_o relative to Hα, then

$$I_{\mathrm{H}\alpha} = Y + L_o, \qquad I_\ell = r_y Y + r_o L_o,$$

$$r_\ell = \frac{I_\ell}{I_{\mathrm{H}\alpha}} = (1-w) r_y + w r_o, \qquad w = \frac{L_o}{Y + L_o}.$$

If Y decreases while L_o stays approximately fixed, w increases. If r_o exceeds r_y, the observed ratio rises even though both Hα and the forbidden line fade. The forbidden-line luminosity declines more slowly because its persistent component has the larger line ratio. No increase in the absolute power of that component is required. This is the key mechanism that can link the Section 4 census to the Section 5 brightness and ratio profiles.

Mix the linear intensities first and take logarithms afterwards. A straight interpolation between logarithmic BPT coordinates is not generally the same model. For [O III]/Hβ, the mixing weight is set by each component's Hβ contribution. It equals the Hα-based weight only under the assumed common Balmer decrement. Different attenuation or intrinsic decrements require a separate Hβ weight and should not be hidden by the plotting convention.

The simple prediction is conditional on r_o > r_y for the particular ratio. A harder source spectrum does not guarantee that every BPT ratio rises by the same amount, or even in the same direction, at arbitrary abundance, ionisation parameter, and gas geometry. Metallicity and N/O are particularly relevant to [N II]/Hα, while [O III]/Hβ also responds strongly to ionisation conditions. Paper I itself motivates care with abundance-related differences across the sample. A changing source mixture is a plausible explanation to be assessed against these alternatives, not a unique inversion from BPT position.

The baseline can also produce declining EW if the stellar continuum changes more slowly than the nebular luminosity. This supplies an additional qualitative connection. It does not predict an exact EW trajectory without a stellar-population and continuum model. The continuum and its age distribution can differ between the stage populations even at the same total stellar surface density.

E3. Ionising photons must still encounter gas

The strongest correction to the physical interpretation is to distinguish the persistence of old stars from the persistence of line luminosity. A useful generalisation of the note's old-star term is

$$L_o = \varepsilon_{\mathrm{H}\alpha} f_{\mathrm{abs},o} \, q_o \, \Sigma_{\mathrm{old}},$$

where q_o is the ionising-photon production rate per unit mass of the relevant evolved population, f_abs,o is the fraction producing the local nebular emission, and epsilon_Hα is the adopted conversion from absorbed ionising photons to Hα luminosity. The note adopts q_o = 7 x 10^40 s^-1 per solar mass and epsilon_Hα = 1.3721 x 10^-12 erg per absorbed photon, giving approximately 9.60 x 10^28 erg/s per solar mass before multiplication by the absorption fraction and the appropriate stellar surface density. This is an adopted normalisation associated with the Belfiore et al. (2022) modelling context, not a MAUVE measurement or a universal constant of all old populations.

If gas is removed, its covering fraction falls, or photons travel beyond the local region, f_abs,o can decrease even when the evolved population is unchanged. Therefore, a persistent stellar photon source does not guarantee a fixed Hα floor. In the innermost retained gas, roughly persistent absorption may be a useful approximation. In the outskirts, the same approximation can fail and line emission may become undetectable. This provides a coherent qualitative connection between the two observed outcomes, but it requires a spatial gas/absorption model to become a quantitative prediction.

The analogous young-star term also depends on absorption and photon transport. At approximately 100 pc scales, ionising stars and the gas that absorbs their radiation need not be co-spatial. The usual conversion between instantaneous local SFR and Hα further assumes that the young population and its recent history are sufficiently well sampled. These issues affect the interpretation of a local residual relative to the rSFMS as well as the physical interpretation of NSF emission.

Recombination emission primarily constrains an emission measure, involving an integral of electron and proton densities along the line of sight. It does not directly measure the remaining ionised-gas mass without a geometry, filling factor, and density distribution. Thus neither a lower surface brightness nor an increased NSF area fraction alone establishes how much gas has been retained or removed.

E4. Leakage, HOLMES, shocks, and mixed explanations

The paper should compare a small set of hypotheses through their expected combinations of observables. In an evolved-star interpretation, the declining young-star contribution reveals a component related to the old stellar population and constrained by its photon budget. In a leakage interpretation, the radiation field depends on nearby H II regions and transport through the disc. In a shock or mixing interpretation, additional heating and gas motions become relevant. Weak nuclear activity, unresolved components, abundance changes, and bulge-dominated stellar populations can also contribute locally. Several mechanisms can coexist.

Belfiore et al. (2022) and the new Baron et al. (2026) preprint provide a particularly useful contrast in how the radiation from young and evolved populations is assessed. Their results should be compared in the star-forming-disc regime actually studied, rather than treated as a single definitive prescription for advanced stripping populations. The recent preprint makes leaked and hardened radiation a substantive alternative. The weak-EW, evolved-population context of Byler et al. (2019) remains relevant where recent SF is small. The MAUVE data can test consistency with these regimes; the present measurements do not uniquely decompose them.

The extra leakage parameter in the 20 October note should describe the fraction of young-star photons absorbed in the diffuse component, for example f_leak,abs. It should not be called a measured galaxy-wide Lyman-continuum escape fraction. Photons escaping an H II region can still be absorbed within the galaxy; photons escaping the galaxy do not produce its local recombination emission. Likewise, a reduction in neutral column does not guarantee monotonically stronger spectral hardening. Filtering, the available photon budget, and absorption all change together and require a radiative-transfer prescription.

The practical tests are correspondingly focused: compare residual Hα with an old-population photon budget and stellar age indicators; compare diffuse emission with nearby H II-region geometry; inspect nuclear and broad-component contributions; and assess line ratios sensitive to partially ionised gas where they are reliably measured. A width excess can motivate shocks, but does not identify them on its own. A sulfur ratio consistent with low density can constrain the emitting phase, but does not establish dilution as the cause of the BPT change.

E5. What the model explains, and what it does not yet predict

The present model explains how a declining young-star contribution can produce lower total line luminosities, lower EW under a slowly changing continuum, and larger selected forbidden-line-to-Balmer ratios under a specified component mixture. These are useful, connected predictions. A spatially varying gas absorption term could link them to inner persistence and outer disappearance, but that extension is not yet computed against the observed selection.

It does not presently predict SF/NSF/ND fractions, fitted BPT classifications, electron densities, or line widths. Predicting those quantities would require spatial distributions, gas structure, radiation or shock calculations, an instrumental response, and the actual detection and classification rules. It also does not measure a stripping timescale or demonstrate that the observed galaxies are successive versions of one initial system. Calling it a physical interpretation or an illustrative model is accurate; calling it a validated evolutionary reconstruction would exceed the evidence.

The source note should be revised before any of its text or equations are transferred to a manuscript. Its Armitage citation has an inconsistent year and points to accretion-disc material that is not the appropriate foundation for the galaxy gas regulator. The galaxy-regulator references above provide the relevant context. There are also duplicated derivative/subscript notation issues in its later equations. No changes were made to that note in this task; the equations presented here use consistent reservoir notation and state the missing absorption assumption explicitly.

F. The analysis needed to turn this framework into the submitted paper

F1. Priorities tied to the main claims

The remaining work should strengthen the connected argument, beginning with the category census. It need not expand into every possible ancillary diagnostic. The following order preserves the central scientific return even if the sulfur or dispersion results remain weak.

Priority and claim	Smallest informative analysis	Consequence for the manuscript
1. Inner NSF versus outer ND	Freeze the common masks; report the usable footprint and detection limits; audit invalid diagnostics in NSF; compare radius and local density jointly with galaxy support.	Establish which spatial contrasts remain under comparable selection and which are only projected trends.
2. A population-stage difference	Repeat the key contrasts after matching or stratifying global stellar mass and bulge/central structure; assess nuclear regions, inclination, and leave-one-galaxy-out sensitivity.	Separate a stage-associated difference from changes in the mix of host galaxies.
3. Fading emission and rising ratios	Compare on a common detectable footprint; examine sensitivity to the individual line cuts, continuum quality, extinction, and limiting surface brightness; test the effect of selection with realistic line uncertainties.	Determine whether differential fading survives changes in the six-line selection.
4. The nearby-galaxy rSFMS comparison	Match resolution, footprint, calibration, binning, weights, and treatment of censored values; identify overlapping galaxies and show an independent-control variant.	Decide whether the result warrants a quantitative comparison or remains contextual.
5. A directional excess in NGC 4654	Harmonise the exploratory and final selections; apply the same sector search to controls and uncertainty resampling; verify sky-angle conventions against the external H I geometry.	Preserve the measured angular result while calibrating the strength of the local-excess and alignment claims.
6. Source-mixture interpretation	Compare absorbed photon budgets, stellar populations, and nearby SF geometry; show one illustrative model with its parameters and assumptions.	Present a physically constrained explanation without claiming a uniquely recovered source mixture.
7. Dispersion and electron density	Check line-fitting ties, instrumental correction, beam smearing, detection reliability, and mass dependence; propagate sulfur flux errors in flux space and retain the low-density plateau.	Include a diagnostic as a principal result only if its trend and sensitivity are demonstrated.

For priority 1, the distinction between gas loss and failure of the Balmer requirement is essential: Hβ can fall below threshold while Hα or forbidden lines remain detected. The report therefore uses "Balmer-ND" descriptively and does not interpret the category as a gas-free region. Similarly, missing BPT information or invalid width/EW measurements can enter the current complement-defined NSF population. A sensitivity analysis restricted to fully measurable diagnostics would reveal how much of the central NSF contrast is attributable to secure emission classification.

For priority 2, local stellar surface density does not automatically control total stellar mass, bulge fraction, age, metallicity, or nuclear activity. These quantities can influence both the probability of NSF classification and its spectrum. A galaxy-level analysis is more informative than treating the millions of pixels as independent evidence. The stage definition should have its catalogue provenance, ambiguities, and combined-galaxy handling recorded in a sample table. If those labels already use H I truncation or SF-related morphology, their relation to the outer SF result also needs to be explained.

For priority 3, requiring all six lines to be detected selects only a subset of NSF locations. The f_NSF census and the spectral-property profiles consequently describe related but different populations. A larger NSF area fraction can coexist with a smaller area meeting the six-line cut. As Balmer emission weakens, denominator-line selection and correlated noise can change the observable ratio distribution. Repeating the profiles under different cuts, with flux-space uncertainty propagation or a forward selection calculation, directly tests the risk. This is a concrete measurement issue rather than a reason to assume that the ratio trend is spurious.

The main figures should retain galaxy counts and identify sparsely supported bins. Whole-galaxy bootstrap intervals, paired contrasts where appropriate, and leave-one-galaxy-out summaries can then show the robustness of the well-supported range. The role of the combined NGC4567_8 product must remain explicit: it is one weighting and resampling unit in the present analysis.

F2. Suggested result wording at the present evidence level

For Section 3: "Within the common, well-sampled stellar-density interval, the stage populations show a systematic decrease in the intensity of the SF-selected component. This population trend coexists with local excesses, including an angular maximum in NGC 4654 within the broad north-west gas-compression sector."

For Section 4: "The decline in SF coverage is accompanied by a spatially dependent change in the emission census. The outer post-peak disc is increasingly represented by regions failing the adopted Balmer-detection criteria, whereas detectable NSF emission is prominent in the inner disc."

For Section 5: "Across much of the supported range, the detectable NSF component has weaker line emission and lower Hα equivalent width in the more processed populations, while several forbidden-line-to-Balmer ratios are larger. The ordering is not universal in every bin, and the spectral-property measurements apply to the subset satisfying the relevant diagnostic requirements."

For Section 6: "A decline in the young-star contribution, accompanied by a more slowly varying contribution from other ionisation sources in retained gas, can account for the joint fading and ratio behaviour. The data constrain this interpretation through the emission census and spectral diagnostics, but do not uniquely identify the ionising source or establish a calibrated evolutionary timescale."

These formulations are provisional manuscript language tied to the saved analyses. Numerical significance, fitted stage offsets, or universal statements should be added only after the corresponding comparison has

been measured. A weak or absent density trend does not invalidate the principal spatial and excitation results.

F3. Relationship to the language and structure of Paper I

The Introduction was calibrated against the published Introduction and framing of Huang et al. (2026). It follows the same broad movement from an observational problem, through resolved measurements and competing interpretations, to a focused MAUVE question and paper roadmap. It uses British scientific spelling and restrained causal language, with references attached to specific observational or modelling claims. The text is newly drafted; it does not reproduce passages from Paper I or assume that a single article establishes every preference of its authors.

The main difference is driven by the science. Paper II requires an explicit connection between three observables: the intensity of remaining SF, the coverage of the emission categories, and the properties of detectable NSF emission. The Introduction therefore gives substantial space to the distinction between SF-selected and full-footprint measurements, and to the established DIG/LIER literature. The individual NGC 4654 paragraph can be shortened in the final manuscript if space is needed, whereas the motivation for the joint census should remain.

G. Source and evidence record

G1. Local material inspected

The following files were inspected in their current local state on 26 October 2026. The notebooks were read as JSON to examine source cells, execution metadata, saved text outputs, and embedded/output-figure context. This does not verify that every saved result was produced from the current inputs; full reproducibility requires execution after the final analysis choices are fixed.

Source	Evidence used here
Main Paper II plotting notebook	Selection logic, estimators, common profile bounds, saved sample counts and regressions, and the figures described in Sections C and D.
Close/post-peak excess screen	Candidate-screen scope and the selection differences between reference and target populations.
NGC 4654 Fourier notebook	Saved directional fit, resampling results, residual summaries, and the coarse H I comparison direction.
20 October physical interpretation note	Two-reservoir equations, illustrative initial conditions, spatial c_HI factor, radiation-mixture assumptions, and limitations to address.
Earlier eight-findings literature report	Context and reference pointers, followed by targeted checks of the relevant primary papers for this report.

Six existing PNG figures in the Paper II figure folder were inspected: rSFMS_PHANGS_MUSE; rSFMS_by_infall_stage; SF_NSF_ND_fractions_by_infall_stage; NSF_emission_line_surface_densities_equal_galaxy; NSF_BPT_ratios_equal_galaxy; and NSF_EW_Hα_NII6583_sigma_SII_doublet_profiles_equal_galaxy. The approximate radial fractions, central doublet values, and curve crossings in this report are visual readings from those saved figures. The Fourier angles and residual statistics instead come from saved numerical outputs.

The main notebook contains saved outputs for its 17 code cells. Its saved valid-population counts are 7,172,265 usable spaxels, comprising 2,364,184 SF, 2,574,225 ND, and 2,233,856 NSF spaxels. These counts add exactly to the usable total, but the pixels are neither spatially independent nor independent draws of galaxies. The fraction profiles and spectral diagnostics apply additional bounds and detection conditions; these totals are not the sample sizes of every plotted point.

No notebook, FITS file, source model note, or existing figure was edited. No raw-product measurements, notebook reruns, new scientific fits, or model parameter exploration were performed. Numerical formula checks are limited to verifying the written reservoir solution against direct integration of its stated equations, including the equal-rate limit; these do not fit or validate the model against MAUVE. The newly created artifacts are this report and its PDF rendering, plus the required concise job-log entry. Rendering and document verification are separate from scientific validation of the underlying analysis.

G2. Literature scope and verification depth

The references below are a focused reading list for the proposed paper, rather than another unrestricted catalogue of relevant publications. Primary publisher pages, author manuscripts, and publication metadata were checked for the claims used here. Relevant full-text passages were inspected for Paper I's style, Pessa's averaging and non-detection treatment, the NGC 4424 interpretation, the closest MAUVE/GASP and LIER comparisons, and the principal DIG and density-diagnostic arguments. Other broad precedents are used at the level supported by their abstracts and publication records. Exact numerical literature comparisons should be rechecked against the final published methods before being adopted as fitted constraints.

The report includes material available by 26 October 2026. Baron et al. (2026) is explicitly a preprint. The bibliography uses journal publication years where those differ from initial preprint or online-first dates; in particular, the cited Belfiore and Lacerda papers are the 2017 and 2018 journal papers. A supplementary Crossref metadata query was rate-limited for some entries; their primary manuscript/publisher records were already checked, and no bibliographic fact is inferred from a failed query. This is a targeted literature assessment, not a systematic review or proof of absolute priority.

H. Focused bibliography

The 38 entries below support the Introduction, the nearest-precedent comparison, and the physical discussion. Author names and typography are transliterated to ASCII in the Markdown source. Publication links provide the authoritative records for final bibliography preparation.

Barnes et al. (2021). *Comparing the pre-SNe feedback and environmental pressures for 6000 H II regions across 19 nearby spiral galaxies.* MNRAS, 508, 5362. Published article; author manuscript. Role: pressure and density interpretation, including emission weighting.

Baron et al. (2026). *DIGging into H II regions: the role of ionizing photon leakage and spectral hardening in coupling H II regions and diffuse ionized gas.* arXiv:2610.07163; full manuscript. Status: preprint submitted on 5 October 2026. Role: current leakage/hardening alternative, particularly for actively star-forming discs.

Belfiore et al. (2016). *SDSS IV MaNGA - spatially resolved diagnostic diagrams: a proof that many galaxies are LIERs.* MNRAS, 461, 3111. Published article. Role: extended low-ionisation emission and distributed ionising sources.

Belfiore et al. (2017). *SDSS-IV MaNGA - the spatially resolved transition from star formation to quiescence.* MNRAS, 466, 2570. Published article; author manuscript. Role: central/extended LIER configurations and the established residual-emission precedent.

Belfiore et al. (2022). *A tale of two DIGs: The relative role of H II regions and low-mass hot evolved stars in powering the diffuse ionised gas (DIG) in PHANGS-MUSE galaxies.* A&A, 659, A26. Published article; full manuscript. Role: leakage, evolved stars, and ionising-photon budgets.

Boselli et al. (2014). *Cold gas properties of the Herschel Reference Survey. III. Molecular gas stripping in cluster galaxies.* A&A, 564, A67. Published article; author manuscript. Role: the limitations of a model neglecting direct environmental effects on molecular gas.

Boselli et al. (2018). *A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE). IV. A tail of ionised gas in the merger remnant NGC 4424.* A&A, 620, A164. Published article; author manuscript. Role: central starburst, outflow, and mixed environmental history.

Boselli et al. (2020). *A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE). VI. Environmental quenching on H II region scales.* A&A, 634, L1. Published article; author manuscript. Role: the closest high-resolution precedent for the loss of outer SF regions.

Boselli, Fossati & Sun (2022). *Ram pressure stripping in high-density environments.* A&A Review, 30, 3. Published review; author manuscript. Role: multiphase environmental framework.

Brown et al. (2023). *VERTICO. VII. Environmental Quenching Caused by the Suppression of Molecular Gas Content and Star Formation Efficiency in Virgo Cluster Galaxies.* ApJ, 956, 37. Published article; author manuscript. Role: direct Virgo resolved-SF and gas-cycle comparison.

Brown et al. (2026). *MAUVE-MUSE: Ionization and Kinematic Signatures of Environmental Effects on Virgo Cluster Disks.* ApJ, 1004, 73. Published article; full manuscript. Role: the closest prior MAUVE excitation and kinematics study.

Byler et al. (2019). *Self-consistent Predictions for LIER-like Emission Lines from Post-AGB Stars.* AJ, 158, 2. Published article; author manuscript. Role: an evolved-population photoionisation model and weak-EW expectations.

Cakir et al. (2026). *The SAMI Galaxy Survey: Quenching of Star Formation in Clusters III. Ram-Pressure-Affected Galaxy Populations.* PASA, 43, e029. Published article; author manuscript. Role: recent resolved stage/morphology comparison and outer suppression.

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